

Write a program to add 2 numbers 05h and 03H

CODE :

```
ORG 000H
```

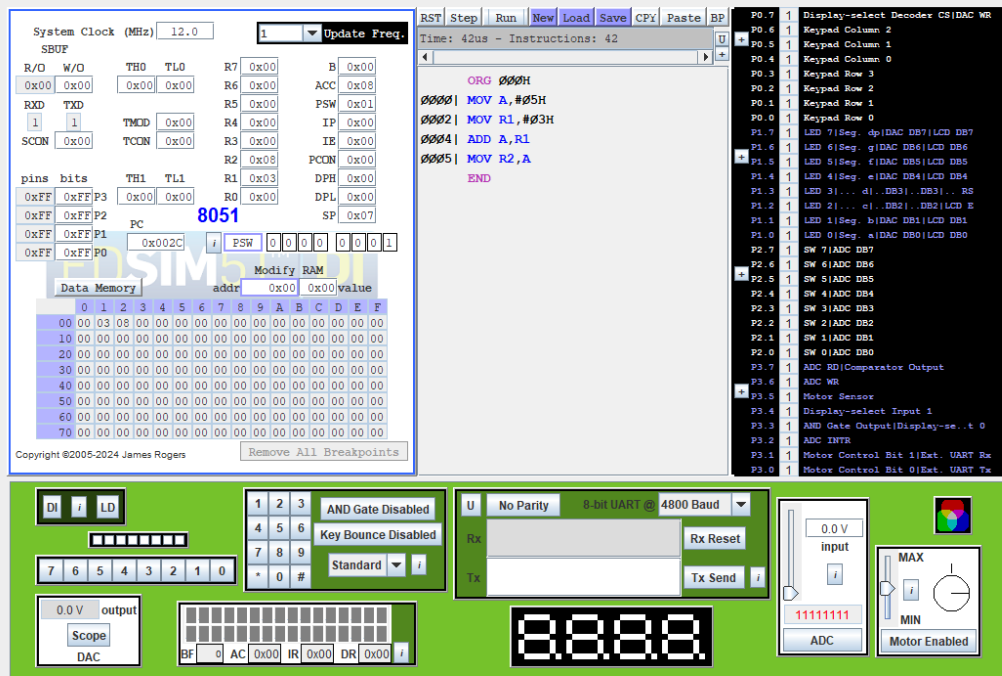
```
MOV A,#05H
```

```
MOV R1,#03H
```

```
ADD A,R1
```

```
MOV R2,A
```

```
END
```



Write a program to SUBTRACT 2 numbers 05h and 03H

CODE

ORG 000H

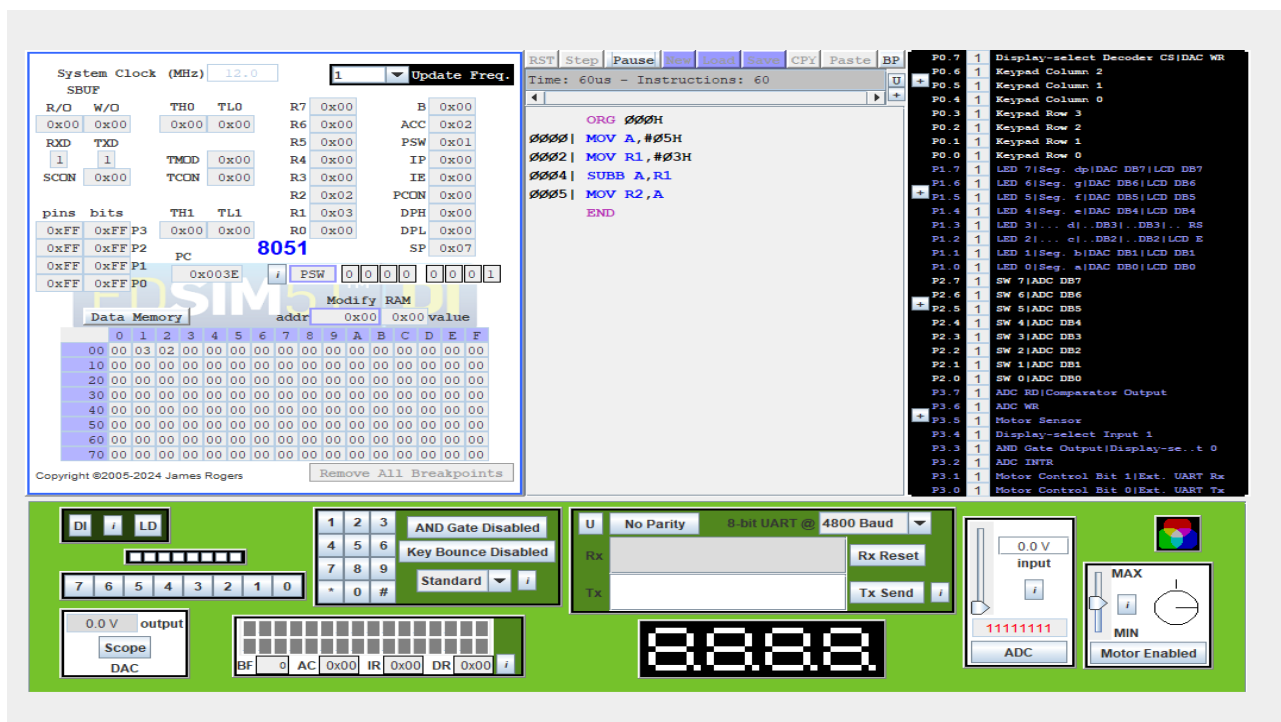
MOV A,#05H

MOV R1,#03H

SUBB A,R1

MOV R2,A

END



WRITE A PROGRAM TO MULTIPLY 2 NUM

CODE :

ORG 000H

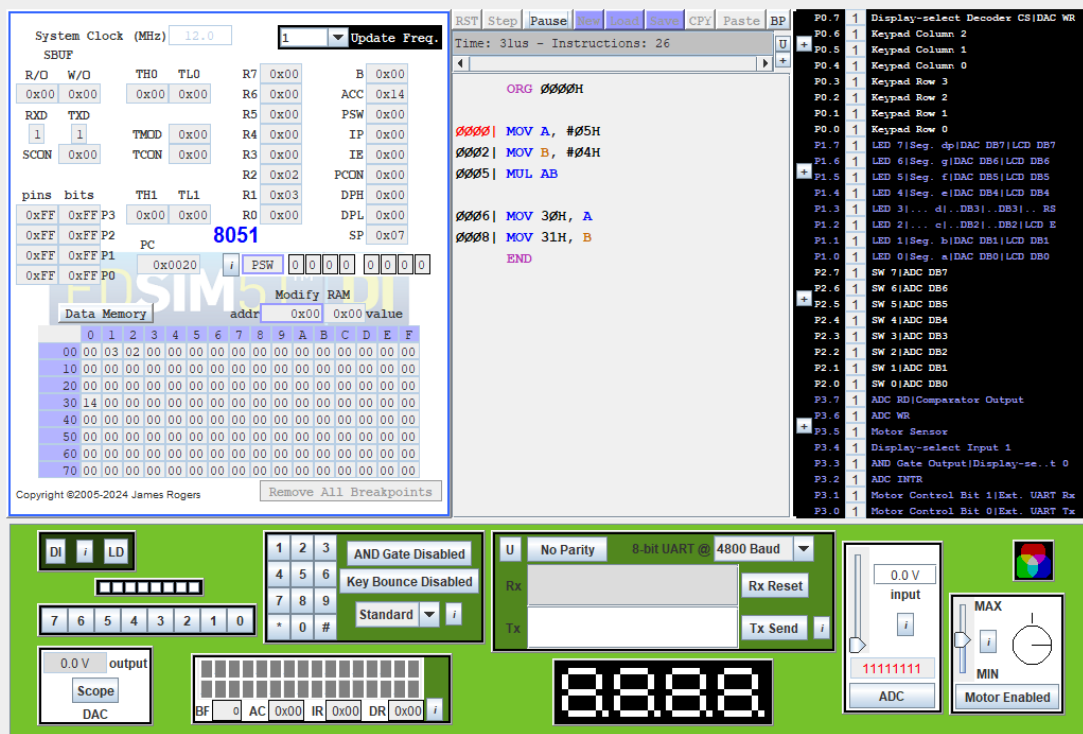
MOV A,#05H

MOV B,#04H

MUL AB

MOV R2,A

END



WRITE A PROGRAM TO DIVIDE 10 H WITH 03H AND SAVE THE QUOTIENT IN R4
AND REMAINDER IN R5

CODE :

ORG 000H

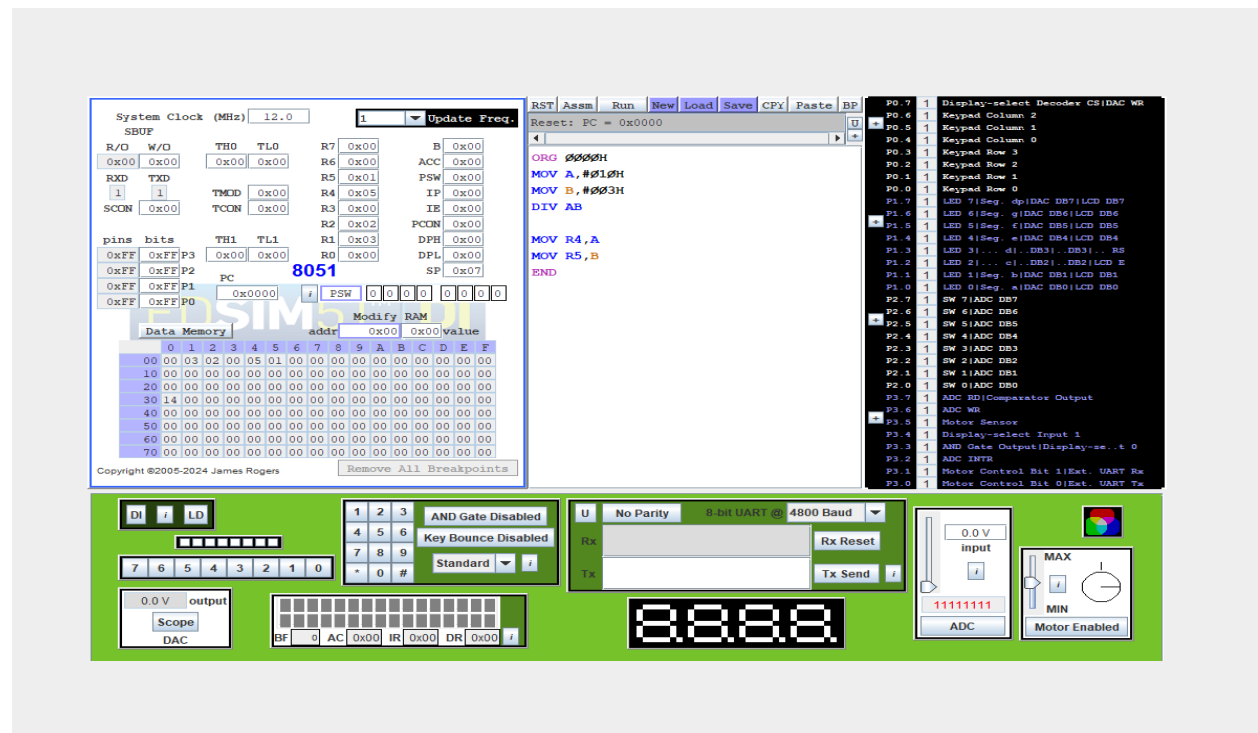
MOV A,#08H

MOV B,#03H

DIV AB

MOV R2,A

END



Write a program to swap 2 numbers ,55H to be stored in R0,AAH stores in R1 , swap r0
and r1

CODE:

ORG 000H

MOV R0,#55H

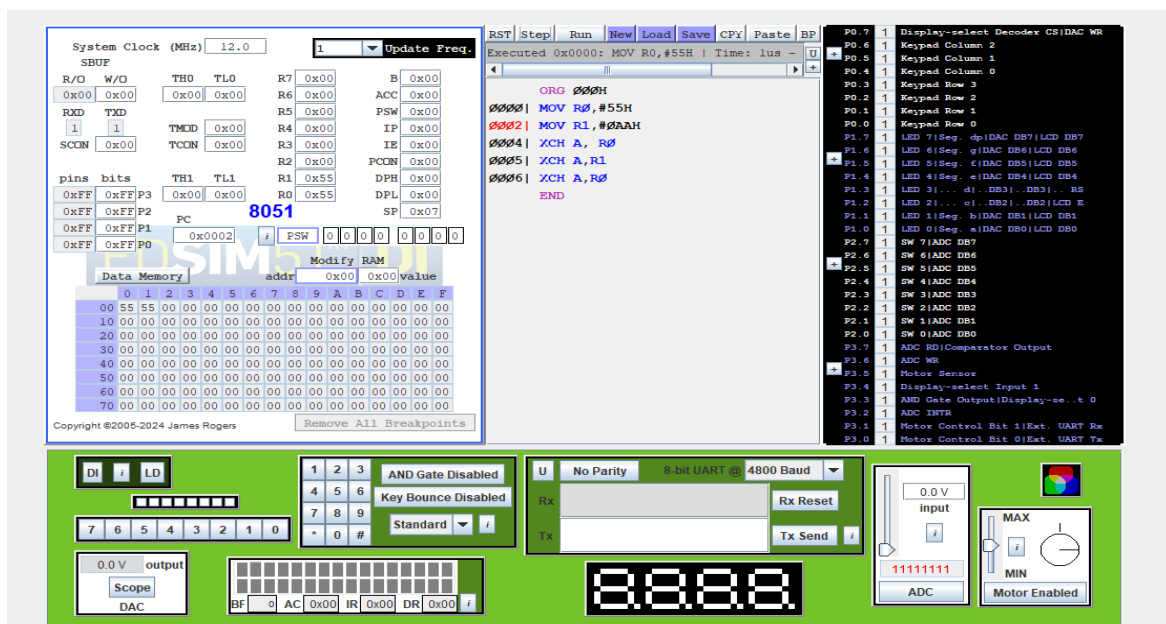
MOV R1,#0AAH

MOV A,R0

XCH A,R1

MOV R0,A

END



WRITE A PROGRAM TO BLINK LED ON PORT 1

CODE :

ORG 000H

SETB P1.0

BLINK:

CPL P1.0

ACALL DELAY

SJMP BLINK

DELAY:

MOV R7,#0FFH

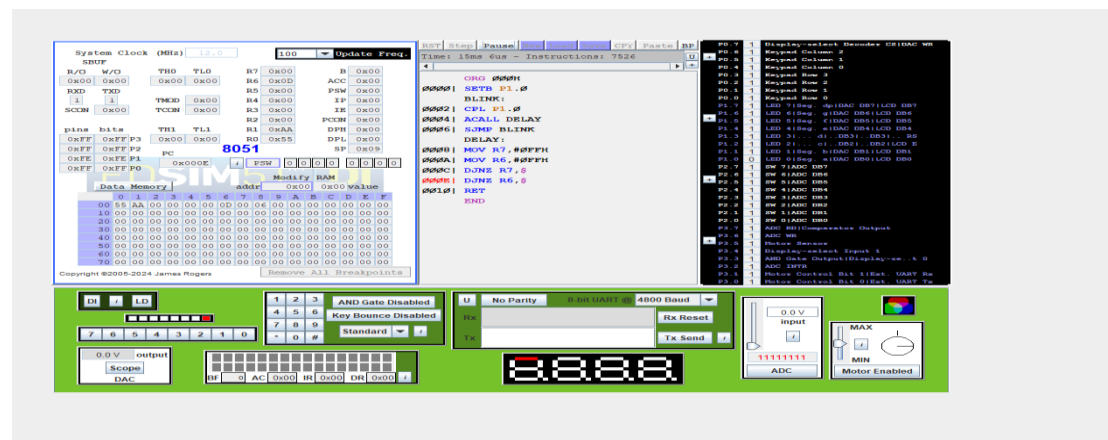
MOV R6,#0FFH

DJNZ R7,\$

DJNZ R6,\$

RET

END



WRITE A PROGRAM TO BINK LED RIGHT TO LEFT

CODE:

START:

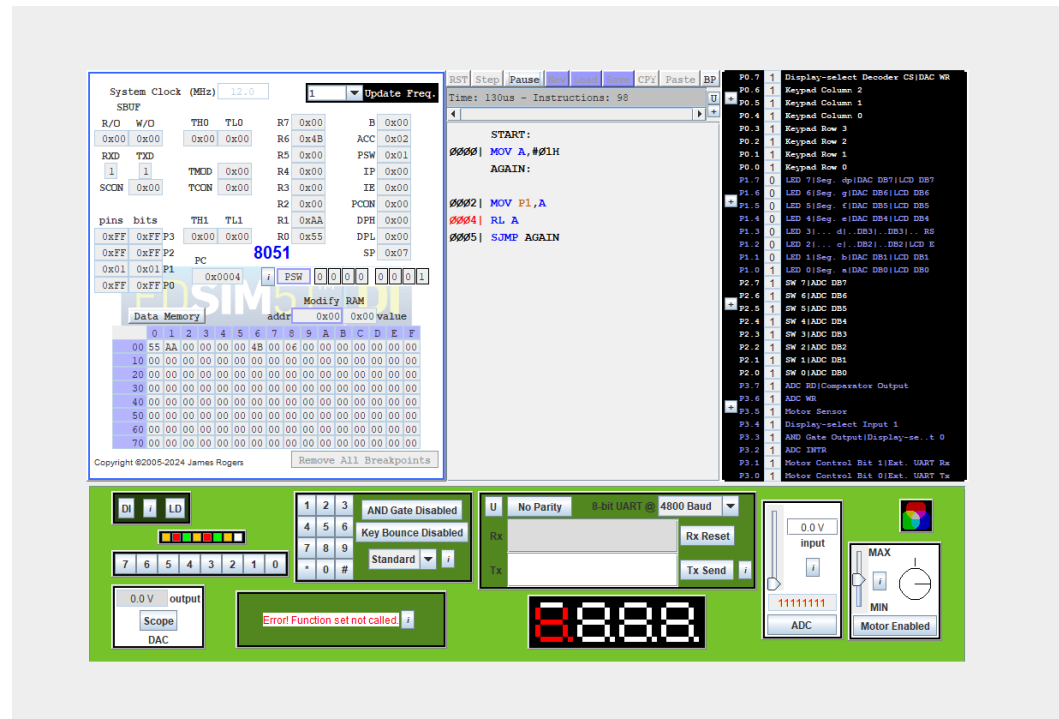
MOV A,#01H

AGAIN:

MOV P1,A

RL A

SJMP AGAIN



TO PRINT 8051 IN THE DISPLAY

ORG 0000H

START:

MOV P1,#7FH ; 8

```

ACALL DELAY
MOV P1,#3FH ; 0
ACALL DELAY
MOV P1,#6DH ; 5
ACALL DELAY
MOV P1,#06H ; 1
ACALL DELAY
SJMP START

```

DELAY:

```
MOV R6,#255
```

```
D1: MOV R7,#255
```

```
D2: DJNZ R7,D2
```

```
DJNZ R6,D1
```

```
RET
```

```
END
```

The screenshot displays the Proteus 8051 simulator interface. The main window shows the assembly code being executed:

```

0000| MOV P1,#7FH ; 8
0003| ACALL DELAY
0005| MOV P1,#3FH ; 0
0008| ACALL DELAY
000A| MOV P1,#6DH ; 5
000D| ACALL DELAY
000F| MOV P1,#06H ; 1
0012| ACALL DELAY
0014| SJMP START

```

Below the code, the **DELAY:** section is visible:

```

0016| MOV R6,#255
0018| D1: MOV R7,#255
001A| D2: DJNZ R7,D2
001C| DJNZ R6,D1
001E| RET

```

The left panel shows the register file with R7 at 0x5E and R6 at 0xFF. The PC is at 0x001A. The Data Memory window shows the current memory state. The bottom panel features a hardware interface with a keypad, a 7-segment display showing '8888', an ADC input set to 0.0V, and a motor control section.